

SUPPLEMENTARY SEMESTER EXAMINATIONS - SEPTEMBER 2022

Program	: B.E. : Computer Science and Engineering	Semester	: III
Course Name	: Data Structures	Max. Marks	: 100
Course Code	: CS32	Duration	: 3 Hrs

Instructions to the Candidates:

- Answer one full question from each unit.

UNIT- I

- Define algorithm. Explain the various criteria of an algorithm. CO1 (07)
 - Justify the statements "pointers can be dangerous" with an example. CO1 (05)
 - Write a program to sort the given numbers using selection sort algorithm and also define the macros to swap two numbers. CO1 (08)
- Define the following with an example: CO1 (06)
 - Void *
 - Dangling Reference
 - Abstract Data Type.
 - Write an ADT for the following: CO1 (08)
 - Arrays
 - Strings.
 - With the help of example, bring out the two differences between the following: CO1 (06)
 - Static and dynamic memory allocation
 - Structures, unions and self-referential structures.

UNIT - II

- Write an algorithm to evaluate postfix expression and use the same to evaluate the following expression $7\ 2\ 3\ * -\ 8\ 6\ 2\ -\ /\ *$ CO2 (06)
 - Write an ADT for a queue and include necessary operations. CO2 (06)
 - Represent two stacks in an array, `memory[MEMORY_SIZE]`. Write C functions to perform push and pop operations on these stacks. CO2 (08)
- Write the algorithm to convert an infix expression to its postfix equivalent and apply the same to convert the following infix expression to postfix expression:
 $A+(B-C*D)-E+F/G$ CO2 (08)
 - Illustrate add and delete operations on a circular queue with suitable examples. CO2 (06)
 - Illustrate push and pop operations on multiple stacks with a neat diagram. CO2 (06)

UNIT - III

- Write a C function to concatenate two singly linked list. CO3 (06)
 - Bring out the differences between the following: CO3 (04)
 - Singly linked list and Doubly linked list
 - Circular singly linked list and circular doubly linked list.
 - Write a program to implement stacks using singly linked list with header nodes. CO3 (10)

6. a) Write a C function to delete from a doubly linked circular list. CO3 (06)
b) List any two advantages and disadvantages of doubly linked list. CO3 (04)
c) Write a program to implement multiple queues represented as a linked list. CO3 (10)

UNIT – IV

7. a) Define binary tree. Write any three properties of binary trees. CO4 (07)
b) Write the structure definition for threaded binary tree. Show how will be the insertion of new node happen? CO4 (08)
c) What are winner trees and loser trees? Give examples and brief about them. CO4 (05)
8. a) Show the tracing of in-order traversal of binary tree with an example. CO4 (07)
b) Write a C function to perform deletion from Max heap. CO4 (08)
c) How will you construct a binary tree from its in-order and pre-order sequences? CO4 (05)

UNIT – V

9. a) How will the graph can be represented using adjacency multilists? Show with an example. CO5 (07)
b) What are height biased leftist trees and weight biased leftist trees? Give examples. CO5 (08)
c) Differentiate single and double ended priority queues. CO5 (05)
10. a) Define ADT for Graph with the following operations : CO5 (07)
i) Insertion and deletion of vertices, edges
ii) Checking graph is empty or not
iii) Printing adjacent vertices of node.
b) Write two C functions to perform depth first search and breadth first search of a graph. CO5 (08)
c) How will be the balance factor calculated in AVL tree? How is rebalancing happen using different rotations in AVL tree? Show examples. CO5 (05)
